Manual for *RHEED Simulator*

by Chong LIU

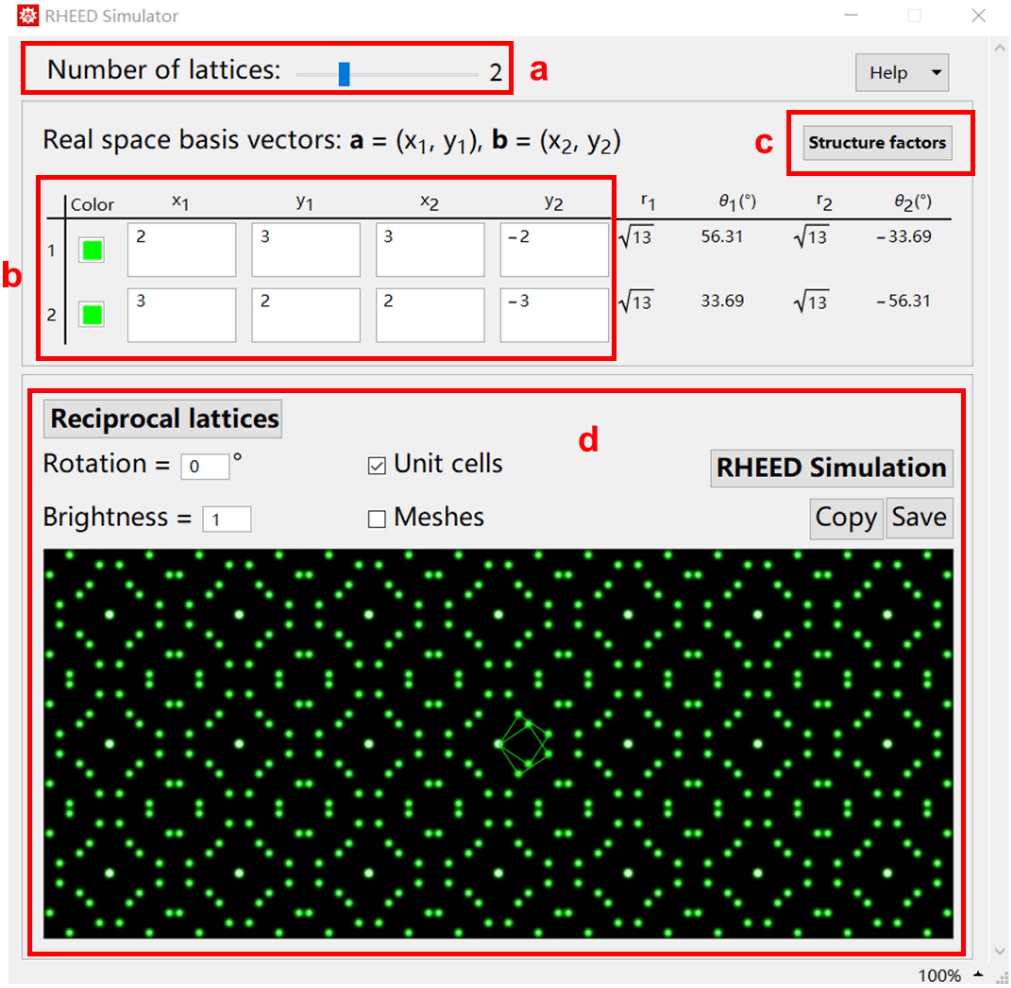

Fig.1 Structure construction interface.

1. Wolfram Mathematica should be installed to use the program. The full program is contained in *RHEED Simulator.cdf*. To run it, double-click on the file or open it in the Mathematica menu *File* > *Open*.

2. Once running the *RHEED Simulator.cdf* file, it opens up a window as shown in Fig.1. The interface mainly contains two panels, the structure setup in real space (a-c) and the preview of reciprocal lattice (d).

3. Set the number of independent lattices in (a). The slider maximum is 5, beyond which one can directly edit the number following, putting in any positive integers. The number of rows of table (b) updates automatically when (a) is changed.

4. In (b), each row defines the coordinates of the basis vectors **a** = (x_1, y_1), **b** = (x_2, y_2). The values are in arbitrary unit here. I recommend use 1 for the unreconstructed unit cell. The absolute length unit of the lattice constants can be

given later on. The (r, θ) columns show the length and angle of the vectors for double-checking. In the example here, it tells the two unit cells are $\sqrt{13} \times \sqrt{13}$ and are rotated by $33.69°$ and $-33.69°$, respectively.

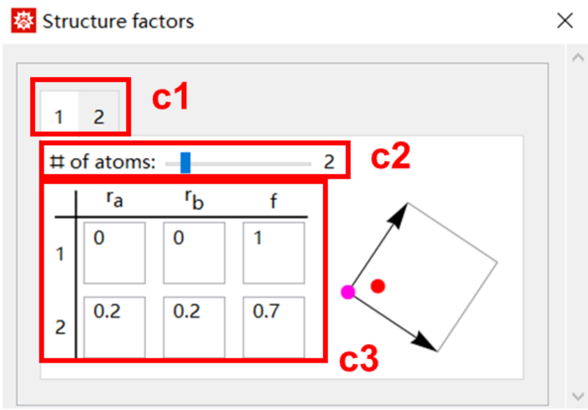

Fig.2 Structure factor setup.

5. Click *Structure factors* button (c). It opens up a new window as shown in Fig.2. The tab numbers (c1) are corresponding to the lattice numbers in (b). Set the number of atoms within the unit cell (c2) (by default it is 1). Set the relative coordinates (r_a,r_b) (0~1) and relative scattering factor of each atom in (c3). The graph previews the unit cell and positions of the atoms. (The parameters in Fig.2 are just for purpose of demonstration, irrelevant to $\sqrt{13} \times \sqrt{13}$ simulation.)

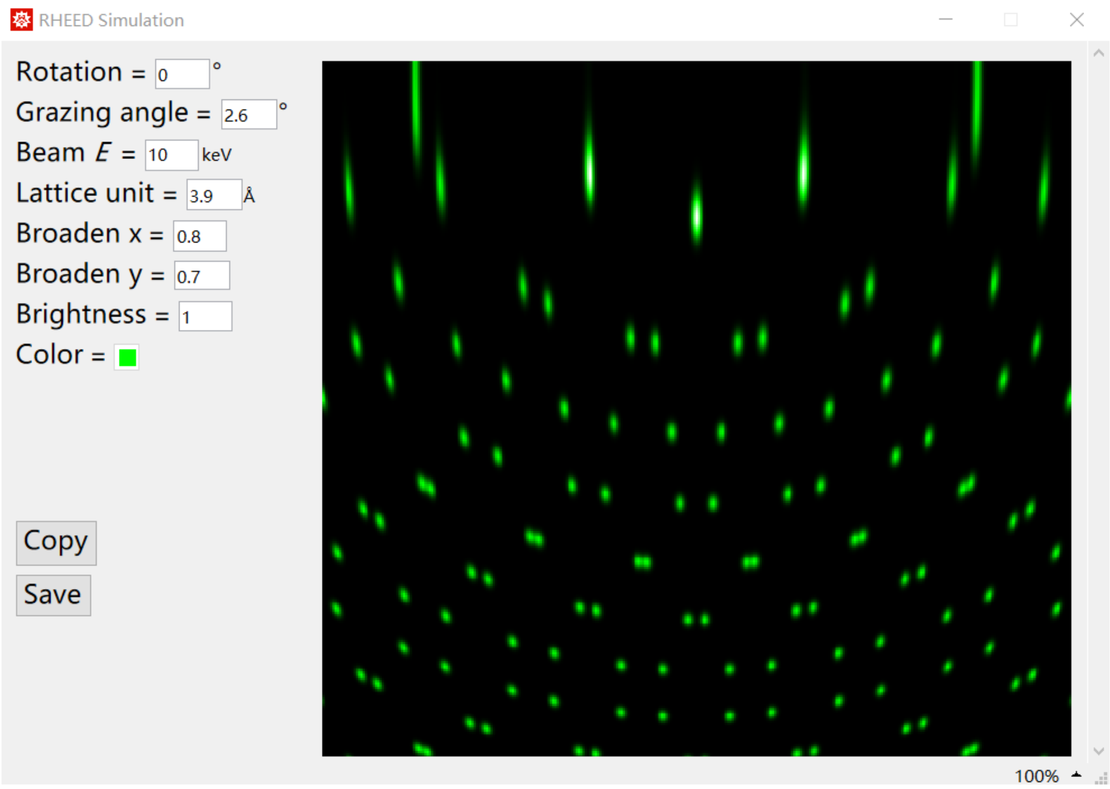

Fig.3 RHEED simulation interface.

6. Back to the main window in Fig.1. In (d), click *Reciprocal lattices* button then the

pattern is updated, which can also be viewed as a simulation of LEED approximately. One can check *Unit cells* or *Meshes* to see which lattice the spots belong to. The *Rotation* can rotate the whole pattern. The rotating angle will be passed on to the initial RHEED simulation.

7. Click *RHEED Simulation* button. A new window will be created as shown in Fig.3. The simulated RHEED pattern is displayed on the right. Customize the parameters:

   Rotation: any angle desired.

   Grazing angle: 1~6 degree is a reasonable range.

   Beam *E*: 10~15 keV.

   Lattice unit: 1~20 Å. Unit length corresponds to 1 in (b).

   Broaden x & Broaden y: 0.2~5. Broadening factors of spots along x and y directions.

   Brightness: Overall intensity rescaling.

   Color: Open up a color setter to pick up the color of the spots.

   The simulation is automatically updated after any parameter is changed.

8. Click *Copy* button to copy the graph to clipboard. Clicking *Save* button will bring up a dialog window to select a local directory and set the file name. This will save the graph in common image formats.

   To save the model parameters in the main interface in Fig.1, pressing Ctrl+s will save in the current file and Ctrl+Shift+s will save as a new file.

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